

We have shown that:

$$T(X_s) = \text{diag}((\gamma^* \mathbf{1}_{n_t})^{-1}) \gamma^* X_t$$

We assume that the weights are uniform:

$$p_i^s = \frac{1}{n_s} \quad \forall i = 1, \dots, n_s$$

$$p_j^t = \frac{1}{n_t} \quad \forall j = 1, \dots, n_t$$

Therefore:

$$\begin{aligned} \gamma^* \mathbf{1}_{n_t} &= \begin{pmatrix} p_1^s \\ \vdots \\ p_{n_s}^s \end{pmatrix} = \begin{pmatrix} \frac{1}{n_s} \\ \vdots \\ \frac{1}{n_s} \end{pmatrix} = \frac{1}{n_s} \mathbf{1}_{n_s} \\ \text{diag}(\gamma^* \mathbf{1}_{n_t}) &= \text{diag}\left(\frac{1}{n_s}, \dots, \frac{1}{n_s}\right) = \frac{1}{n_s} I_{n_s} \end{aligned}$$

Thus:

$$\text{diag}(\gamma^* \mathbf{1}_{n_t})^{-1} = n_s I_{n_s}$$

Consequently:

$$T(X_s) = \text{diag}(\gamma^* \mathbf{1}_{n_t})^{-1} \gamma^* X_t = n_s I_{n_s} \gamma^* X_t = n_s \gamma^* X_t$$